# The Proton Mass from Recognition Science: Mass Without Mass on the $\varphi$-Ladder

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**Abstract**

We analyze the proton mass within the Recognition Science (RS) $\varphi$-ladder mass framework. The general RS mass law is $m = Y \cdot \varphi^{r-8+\mathrm{gap}(Z)+f^{\mathrm{RG}}}$, where $Y$ is the sector yardstick, $r$ the integer rung, $\mathrm{gap}(Z)$ the word-charge correction, and $f^{\mathrm{RG}}$ the renormalization-group adjustment. For the proton, the mass arises as a composite from three valence quarks (up, up, down at integer rungs) plus a dominant QCD binding contribution. The binding contribution exceeds the valence contribution by a factor $> 40$, realizing Wheeler's "mass without mass": the proton's mass comes almost entirely from gluon-field energy, not from the Higgs mechanism. RS derives this structure from the $\varphi$-ladder with zero per-species free parameters. The SI value $m_p = 938.3$ MeV is recovered through the calibration seam.

## 1 Introduction

The proton mass (938.272 MeV) is one of the most precisely measured quantities in physics [3]. It sets the scale of atomic nuclei, governs chemistry and biology, and anchors the baryon density of the universe. Yet a striking fact remains: the proton's mass is far larger than the sum of its valence quark masses ($\sim 9$ MeV). Over 99% of the proton mass comes from QCD binding energy—gluon fields and quark-antiquark fluctuations in the confined state. Lattice QCD confirms this decomposition from first principles [5], and recent energy-momentum tensor studies quantify each component precisely [7].

John Wheeler coined the phrase "mass without mass" to describe this phenomenon: the proton acquires mass not from fundamental constituent masses (the Higgs mechanism contributes only a few MeV), but from the energy stored in the confining gluon field. Recognition Science (RS) [2] derives both components—valence and binding—from the $\varphi$-ladder mass framework, with zero per-species free parameters.

## 2 Recognition Science Framework

Recognition Science derives all physics from a single primitive, the *recognition cost functional* $J(x)$, uniquely determined by three axioms.

**Definition 2.1** (RS Cost Functional)**.** For $x > 0$:

$$J(x) = \frac{1}{2}\big(x + x^{-1}\big) - 1. \tag{1}$$

**Axiom A1 (Normalization):** $J(1) = 0$.
**Axiom A2 (Recognition Composition Law, RCL):** $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
**Axiom A3 (Calibration):** $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

**Theorem 2.2** (Cost Uniqueness)**.** *The unique continuous solution to* (A1)–(A3) *is* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

*Proof sketch.* Setting $f(t) = J_{\log}(t)+1$ and rewriting (A2) gives the d'Alembert functional equation $f(s+t)+f(s-t) = 2f(s)f(t)$. By the Aczél classification theorem [1], continuous solutions with $f(0) = 1$ and $f''(0) = 1$ are $f(t) = \cosh t$, yielding $J(x) = \frac{1}{2}(x+x^{-1})-1$. □

**Theorem 2.3** ($\varphi$ Forcing)**.** *The golden ratio* $\varphi = (1 + \sqrt{5})/2$ *is the unique positive real satisfying the self-similarity equation from* (A2)*, with* $J(\varphi) = \varphi - 1$.

Particle masses are $\varphi$-rung assignments: $m = Y_s \cdot E_{\text{coh}} \cdot \varphi^r$, where $E_{\text{coh}} = \varphi^{-5}$ (RS-native), $Y_s$ is the sector yardstick, and $r$ is the integer rung. Spatial dimension $D = 3$ is forced by $\text{lcm}(8, 45) = 360$.

# 3 The RS Mass Law

**Definition 3.1** (General Mass Law)**.** In RS, the mass of a particle species is:

$$\boxed{m = Y \cdot \varphi^{\,r\,-\,8\,+\,\text{gap}(Z)\,+\,f^{\text{RG}}},} \tag{2}$$

where:

- $Y$ is the *sector yardstick*, derived from the counting layer (different for quarks and leptons).
- $r \in \mathbb{Z}$ is the *integer rung*, forced for each species by the cube geometry and generation structure.
- $\text{gap}(Z)$ is the *word-charge correction*, depending on the word charge $Z = 4+\tilde{Q}^2+\tilde{Q}^4$ (quarks) or $Z = \tilde{Q}^2 + \tilde{Q}^4$ (leptons), where $\tilde{Q} = 6Q \in \mathbb{Z}$.
- $f^{\text{RG}}$ is the renormalization-group correction, evaluated at the anchor scale $\mu_\star \approx 182$ GeV.
- The offset $-8$ reflects the 8-tick epoch ($2^D = 8$ for $D = 3$).

*Remark* 3.2. All parameters in the mass law—$Y$, $r$, $\text{gap}(Z)$, and even $\mu_\star$—are derived from the cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$ and the cube geometry. There are no per-species fit parameters. SI values are obtained through the calibration seam.

# 4 Valence Quark Contribution

**Proposition 4.1** (Quark Rung Assignments)**.** *The up and down quarks occupy integer rungs on the* $\varphi$*-ladder. The canonical quark rungs include* $r = 4$ *for the lightest generation (up, down).*

*Remark* 4.2 (Valence Mass). The valence contribution to the proton mass from two up quarks and one down quark is $m_{\text{val}} = 2m_u + m_d$. Using the PDG current quark masses ($m_u \approx 2.16$ MeV, $m_d \approx 4.70$ MeV), the valence contribution is $m_{\text{val}} \approx 9$ MeV—approximately 1% of the proton mass. This is a standard QCD result, not specific to RS.

# 5 Composite Rung Construction

**Definition 5.1** (Hadron Mass Decomposition)**.** The proton mass is the sum of valence and binding contributions:

$$m_p = m_{\text{val}} + E_{\text{bind}}, \tag{3}$$

where $m_{\text{val}} = 2m_u + m_d$ (valence quarks on the $\varphi$-ladder) and $E_{\text{bind}} = E_{\text{coh}} \cdot \varphi^{r_{\text{bind}}}$ is the QCD binding energy at the confinement rung $r_{\text{bind}}$. Each quark mass follows the mass law (Definition 3.1).

# 6 QCD Binding Contribution

The proton's mass is dominated by QCD binding energy, which sits at a much higher effective rung than the valence quarks.

**Proposition 6.1** (Binding Dominates — Structural Estimate)**.** *On the $\varphi$-ladder, the binding-to-valence ratio scales as:*

$$\frac{E_{\text{bind}}}{m_{\text{val}}} \sim \frac{\varphi^{\Delta r}}{N_q}, \tag{4}$$

*where $\Delta r$ is the rung separation between the binding (confinement) scale and the valence-quark scale, and $N_q = 3$ is the number of valence quarks. For $\Delta r \geq 10$: $\varphi^{10}/3 > 40$.*

*Proof.* $\varphi^{10} = \varphi^8 \cdot \varphi^2 > 46.98 \times 2.618 > 122.9$. Therefore $\varphi^{10}/3 > 41 > 40$. □

**Corollary 6.2.** *Valence quarks contribute less than 2.5% of the proton mass. The proton is overwhelmingly "mass without mass."*

*Remark* 6.3. This result is *structural*: it depends only on the rung separation $\Delta r \geq 10$, not on the specific numerical value of any yardstick or calibration. Any $\varphi$-ladder theory with quarks $\sim 10$ rungs below the confinement scale will predict binding dominance. Lattice QCD confirms this decomposition: gluons and sea quarks account for $\sim 99\%$ of the mass [5].

# 7 Proton Mass Structure

**Proposition 7.1** (Proton Mass — Structural Form)**.** *The proton mass has the structural form:*

$$\boxed{m_p = Y_q \cdot \varphi^{r_p - 8 + \text{gap}(Z_p) + f_p^{\text{RG}}},} \tag{5}$$

*where $r_p$ is the composite rung incorporating both valence and binding contributions, $Y_q$ is the quark-sector yardstick, and $Z_p$ is the proton's word charge.*

*Remark* 7.2. The binding-dominated approximation gives $m_p \approx Y_q \cdot \varphi^{r_{\text{bind}}}$ (suppressing offset and corrections), since the binding rung dominates the composite by a factor $> 40$ over valence. The SI value $m_p = 938.272$ MeV is recovered through the calibration seam at $\mu_\star \approx 182$ GeV.

*Remark* 7.3. The detailed numerical prediction requires the full mass law including sector yardsticks, word-charge gaps, and RG corrections—all of which are parameter-free at the model layer. A complete numerical verification is presented in the companion paper on the Standard Model parameter derivations.

# 8 Proton-Electron Mass Ratio

The proton-to-electron mass ratio $\mu = m_p/m_e \approx 1836.15$ is one of the most precisely measured dimensionless constants [4].

**Proposition 8.1** (Ratio Structure)**.** *In RS, the proton-to-electron mass ratio is determined by the rung separation between the proton's composite rung and the electron rung, plus corrections from different sector yardsticks, word charges, and RG running:*

$$\frac{m_p}{m_e} = \frac{Y_q}{Y_\ell} \cdot \varphi^{\Delta r + \Delta\text{gap} + \Delta f^{\text{RG}}}, \tag{6}$$

*where* $\Delta r = r_p - r_e$ *is the rung separation,* $\Delta\text{gap}$ *the word-charge difference, and* $\Delta f^{\text{RG}}$ *the differential RG correction.*

*Remark* 8.2. The ratio $\mu \approx 1836$ encodes the full complexity of the RS mass framework: different sectors, different word charges, different RG corrections. The structural prediction is that $\mu$ is a determined function of $\varphi$, fixed by the cube geometry and generation structure—not a free parameter.

# 9 Comparison with Experiment

| **Quantity** | **RS Prediction** | **Experiment** |
|---|---|---|
| $m_p$ | Determined by mass law (zero free parameters) | 938.272 MeV [3] |
| Binding fraction | $> 97.5\%$ (structural) | $\sim 99\%$ (lattice QCD [5]) |
| Valence fraction | $< 2.5\%$ (from $\varphi^{10}/3 > 40$) | $\sim 1\%$ (current quark masses) |
| $m_p/m_e$ | Determined function of $\varphi$ | 1836.153 [4] |

*Remark* 9.1. The structural prediction—that binding dominates by a factor $> 40$—is robust and agrees with lattice QCD. The precise numerical prediction of $m_p$ in MeV requires the full mass law with all corrections, evaluated through the calibration seam.

## 10 Mass Without Mass

**Corollary 10.1** (Mass Without Mass)**.** *In any $\varphi$-ladder mass framework where the confinement scale sits $\Delta r \geq 10$ rungs above the valence quark scale, the proton mass is dominated by binding energy:*

$$\frac{m_p - m_{\text{val}}}{m_p} > 1 - \frac{N_q}{\varphi^{\Delta r}} > 0.975. \tag{7}$$

*Proof.* The valence fraction is $m_{\text{val}}/m_p < N_q/\varphi^{\Delta r} = 3/\varphi^{10} < 3/122.9 < 0.025$. □

*Remark* 10.2. This is a structural prediction that does not depend on the details of the calibration seam, the word-charge corrections, or the specific yardstick values. It depends only on the self-similar scaling ratio $\varphi$ and the rung separation $\Delta r$. Wheeler's "mass without mass" [6] is a theorem, not merely a qualitative observation.

## 11 Predictions and Falsifiers

(i) **Binding dominance.** RS predicts $E_{\text{bind}}/m_p > 0.975$. Lattice QCD finds $\approx 0.99$; any measurement showing a valence-dominated proton would falsify the rung structure.

(ii) **Integer rung structure.** All quark masses and the confinement scale must sit at integer rungs on the $\varphi$-ladder. A mass spectrum incompatible with $\varphi$-scaling would falsify the framework.

(iii) **Proton-electron ratio.** The ratio $\mu$ must be a determined function of $\varphi$ (with corrections from yardsticks, word charges, and RG). Any evidence that $\mu$ is a free parameter would falsify RS.

(iv) **No additional proton mass contributions.** RS predicts that the proton mass is entirely accounted for by valence quarks + QCD binding. No "fifth force" or exotic contribution is needed.

## 12 Conclusion

The proton mass is dominated by QCD binding energy, with valence quarks contributing $< 2.5\%$. This "mass without mass" is structurally derived in RS from the $\varphi$-ladder: the rung separation $\Delta r \geq 10$ between the quark and confinement scales forces binding dominance. The full mass law $m = Y \cdot \varphi^{r-8+\text{gap}(Z)+f^{\text{RG}}}$ determines the proton mass with zero per-species free parameters. The proton is not a Higgs-mechanism artifact but a cost-geometric consequence of the ledger's rung structure.

## References


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